

STUDY YKP3089C016

This study is “An Open-Label, Pharmacokinetic Interaction Study Evaluating the Reciprocal Effect of YKP3089 and Phenytoin Following Oral Administration in Healthy Adult Volunteers”.

1. Subject Number 4-17 in Phenytoin Interaction Study

A 19 year old healthy female volunteer subject developed fever, rash, elevated hepatic enzymes, glossitis, and blisters on the lips. The symptoms began 15 days after starting phenytoin and after a single dose of YKP3089. The signs and symptoms are consistent with an antiepileptic hypersensitivity syndrome that has been well-described in patients taking phenytoin.

The Registry of Severe Cutaneous Adverse Reactions (RegiSCAR) score indicates no Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) syndrome for this subject. SKLSI believes causality is not likely related due to YKP3089 because symptoms began immediately after the first dose of YKP3089.

STUDY YKP3089C017

This study is “A Multicenter, Double blind, Randomized, Placebo-Controlled, Dose Response Trial of YKP3089 as Adjunctive Therapy in Subjects with Partial Onset Seizures with Optional Open-Labelled Extension”.

2. Subject Number 01029-001 in YKP3089C017 Study

A 36 year old female with partial epilepsy was enrolled in this study. The subject was taking zonisamide, levetiracetam, and valproic acid when randomized to the 200 mg/day YKP3089 dose group. The subject began taking YKP3089 at 100 mg/day and was increased by 100 mg per week to a target of 200 mg/day per protocol. The subject developed fever, rash, headache, eosinophilia, lymphadenopathy, pleural effusion, and mildly elevated liver function 21 days after starting YKP3089. YKP3089 was stopped within 48 hours of the development of fever, rash, and headache. The subject recovered within two months with steroid treatment, and then was lost to follow up.

The RegiSCAR score indicates definite DRESS syndrome for this subject. The CIOMS form is included in Appendix 4 SKLSI believes the causality is probably related to YKP3089.

3. Subject Number 01016-006 in YKP3089C017 Study

A 22 year old female, with partial epilepsy since 2011, was enrolled in this study. This subject discontinued lamotrigine in the past due to rash. The subject's concomitant anti-epileptic drugs include clonazepam 0.75 mg, clobazam 40 mg bid, and lacosamide 40 0mg bid.

The subject was randomized to 200 mg/day YKP3089 on 22 April 2014. On 2 May 2014, which was 11 days after starting YKP3089, the subject developed rash and fever. No lymphadenopathy or mucosal involvement was detected. YKP3089 was stopped on 3 May 2014. An early termination visit was held on 5 May 2014. Laboratory findings of 5 May 2015

included ALT 107 (range 10-33 U/L) and AST 96 (range 10-36 U/L). Two days later, on 7 May 2014, a CBC was performed at an outside laboratory. The results showed WBC 2.6 (range 3.8-10.8 Thousands/uL), platelet count 118 (range 140-400 Thousands/uL), and absolute neutrophils 1318 (range 1500-7800 cells/uL). On 28 May 2014 these laboratory-findings had resolved.

The RegiSCAR score indicates no DRESS syndrome for this subject. SKLSI believes causality between the reactions and YKP3089 cannot be ruled out.

4. Subject Number: 11003-001 in YKP3089C017 Study

A 36 year old female with partial epilepsy was enrolled in this study. This subject's concomitant anti-epileptic drugs included valproic acid 1500 mg per day, lacosamide 200 mg bid, and levetiracetam 1500 mg bid.

This subject was randomized on 13 November 2014 to the YKP3089 200 mg/day arm. An adverse event of glandular pharyngitis was reported from 10 November 2014 through 16 November 2014. The subject received trimethoprim-methasoxazole from 12 through 16 November 2014. The subject reported glandular pharyngitis again from 6 December 2014 through 17 December 2014. The subject received amoxicillin clavulanate and metamizole from 6 through 17 December 2014. On 23 December 2014 Liver Function Tests (LFTs) were ALT 127 U/L and AST 165 U/L. The subject was found to have a skin rash on 9 January 2015, 55 days after starting YKP3089. YKP3089 was discontinued on 13 January 2015. The rash resolved on 27 January 2015.

The RegiSCAR score indicates no DRESS syndrome for this subject. This case of rash, accompanied by elevated liver function tests, is complicated by multiple other concomitant drugs. SKLSI believes causality between the reactions and YKP3089 cannot be ruled out.

STUDY YKP3089C018

This study is "A Phase 1, Randomized, Double Blind, Placebo Controlled, Multiple Ascending Dose Study to Evaluate the Safety, Tolerability and Pharmacokinetics of YKP3089 in Healthy Volunteers Using a Titration Regimen"

5. Subject Number 13-81 in Multiple Ascending Dose study

A 50 year old healthy male volunteer was domiciled for this Phase 1 study. The subject was randomized to a target dose of 500 mg/day of YKP3089. Titration began with YKP3089 200 mg/day for Days 1 through 5; titration continued with 300 mg/day for Days 6 through 10, 400 mg/day for Days 11 through 15, and 500 mg/day for another 2 weeks. On Day 14, while receiving 400 mg/day, a diffuse pruritic rash was observed. The subject was closely followed with this rash, for 8 days at the Phase 1 study site; the reason the subject was followed was because the subject was continued to be treated with YKP3089 while the rash was present. This was an attempt to treat through the rash before other symptoms emerged. On Day 22, while on 500 mg/day, the rash spread to the palms and soles, and swelling of the chin and hands was noted. YKP3089 was stopped. On Day 23 the subject developed fever, rash,

leukocytosis, eosinophilia, hepatic enzyme elevation, slightly elevated BUN, and creatinine with electrolyte abnormalities. The subject was hospitalized briefly and treated with steroids and IV fluids. He recovered completely in 4-6 weeks.

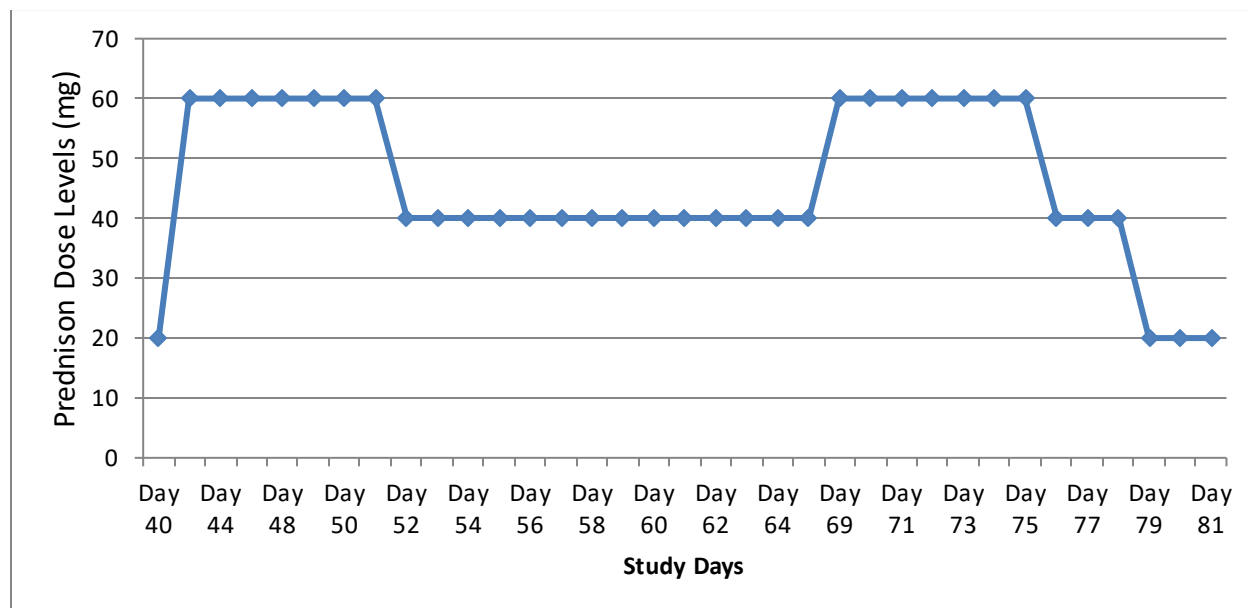
The RegiSCAR score indicates definite DRESS syndrome for this subject. SKLSI believes causality is probably related to YKP3089. See attached CIOMS form for details.

STUDY YKP3089C020

This study is “A Randomized, Double-Blinded, Placebo- and Positive-Controlled, Parallel Study, with a Nested Crossover, Evaluating the Effects of Multiple Oral Dose Administrations of YKP3089 on Electrocardiogram Intervals in Healthy Subjects”, a thorough QT study.

6. Subject Number 95-59 in Thorough QT Study

A 38 year old healthy female volunteer was enrolled in this study. Detailed information can be found in Appendix 4. The subject was randomized to YKP3089. Thirty-two days after starting YKP3089, the subject developed a rash. Approximately 1 week later she developed elevated hepatic enzymes, lymphadenopathy, and eosinophilia. The subject was treated with steroids beginning on day 40 (in below figure). The hepatic enzyme elevations and rash resolved. As a result, prednisone was tapered down over a 1 week period. The subject died seven days after prednisone was stopped.



The RegiSCAR score indicates definite DRESS for this subject. SKLSI believes causality is probably related to YKP3089. See attached CIOMS forms for details.

7. Subject Number 108 in Thorough QT Study

A 26 year old healthy female volunteer was screened for this study on 21 April 2015. The subject did not have any significant medical history. Lab results obtained at screening were WNL, including LFTs, negative virology for HIV, and negative for Hepatitis B and C. The subject had no prior history of rash and no history of reactions to medications.

On 12 May 2015, the subject was randomized and began YKP3089 per protocol titration scheme, with at home dosing beginning Day 3 as follows:

- Day 1-7 50 mg q day
- Day 8-14 100 mg q day
- Day 15-21 150 mg q day
- Day 22-28 200 mg q day

On Day 29, 10 June 2015, the subject developed rash to the trunk and all four proximal extremities, and generalized pruritus. After site evaluation on the same day, the subject was discontinued from study drug. Diphenhydramine was recommended for generalized pruritus. The rash resolved in 2 days. Lab results obtained 10 Jun 2015 showed mildly elevated ALT at 76, mildly decreased HGB at 10.5, HCT at 32.3, PLT count at 155, RBC at 3.64, WBC at 2.5, and Neutrophil count at 1.5. Rechecks on 17 Jun 2015 were ordered. On 17 June 2015 all lab values were WNL except HGB at 10.5, HCT at 31.9, and RBC at 3.61. The ALT was not rechecked on 17 June 2015. On 1 July 2015 the subject complained of parietal headache and edema in bilateral lower legs. On 4 July 2015, the subject complained of burning sensation in bilateral lower legs. On 15 July 2015 the subject complained of edema in bilateral arms and hands. On 27 September 2015 the subject complained of dizziness pain and edema in bilateral upper extremities, pain in bilateral lower extremities, and myalgia. Physical record examination did not confirm these complaints.

The subject was examined by a vascular specialist at the Mayo Clinic on 12 November 2015 as a follow up on the open adverse events of burning sensation bilateral lower legs and edema bilateral lower legs. No signs of vascular problems were identified during the exam. There was no edema in the lower extremities and distal pulses were present and WNL. In the standing position, there was a very mild prominence of one small vein on the lateral aspect of the left leg and near the popliteal area. No evidence of varicosity and no skin discoloration were found.

During a phone call on 18 November 2015 the subject reported that the pain in bilateral upper extremities resolved on 16 November 2015. During a phone call on 8 December 2015, as a follow up on the open adverse events of burning sensation of bilateral lower legs and edema bilateral lower legs, the subject indicated her symptoms continued but were improving. The subject was scheduled to return on 20 December 2015 for a HGB and RBC recheck, and was also referred to an internist. Follow-up is pending.

RegiSCAR score indicates no DRESS syndrome. SKLSI believes causality between the reactions and YKP3089 cannot be ruled out.

STUDY YKP3089C022

This study is "An Open-Label, Multiple-Dose, Pharmacokinetic Drug Interaction Study to Assess the Effects of YKP3089 on Phenobarbital in Healthy Adult Subjects", a phenobarbital interaction study.

8. Subject Number 2-36 in Phenobarbital interaction study

A 52 year old healthy female volunteer was screened for this study on 13 May 2015. The subject had no significant medical history. All lab results obtained at screening were WNL, including hepatic enzymes and antibodies for HIV, Hepatitis B and C. The subject had no prior history of similar rash and no other history of reactions to medications.

On 28 May 2015, the subject began dosing per protocol titration scheme with at home dosing beginning Day 3 as follows:

- Day 1-7 Phenobarbital 30 mg tid day
- Day 8-22 Phenobarbital 90 mg q day
- Day 23-28 Phenobarbital 90 mg q day, YKP3089 50 mg q day
- Day 29-35 Phenobarbital 90 mg q day, YKP3089 100 mg q day
- Day 36-37 Phenobarbital 90 mg q day, YKP3089 150 mg q day

On Day 22, 18 June 2015, during a scheduled visit the subject was found to have a rash on her left cheek. Lab results obtained on the same day, 18 June 2015 (Day 22), showed mildly elevated ALP at 140 U/L, ALT at 73 U/L, slightly decreased WBC at 3.8, eosinophil counts at 1.4 Thou/uL, neutrophil at 37.4 %, and slightly elevated lymphocyte at 52.3 %. The lab tests were repeated on 21 June 2015, Day 25. These lab results showed similar results with mildly elevated ALP at 134 U/L, ALT at 64 U/L, minimally decreased neutrophil at 41.8 %, and slightly elevated lymphocyte at 47.2 %.

On Day 37 (03 July 2015), the subject developed angioedema with urticaria and bilateral preauricular lymphadenopathy. Dosing of both phenobarbital and YKP3089 was stopped on Day 38, 4 July 2015. Prednisone 60mg was started on 4 July 2015 and tapered to 40mg by 7 July 2015. The lab results obtained on Day 39, 5 July 2015, showed mildly elevated ALP at 156 U/L, GGT at 111 U/L, and monocytes at 13.1 %. On 8 July 2015, the subject improved, however, the left preauricular lymph node continued to be enlarged and mildly tender to palpation. The subject continued on prednisone taper until prednisone was stopped on 12 July 2015. The subject was treated again briefly with prednisone from 17 Aug 2015 to 21 Aug for angioedema and urticaria. The subject was evaluated by a Mayo Clinic allergist and advised to take Zyrtec for two months for urticaria. The subject continues to be seen by an allergist and dermatologist for the cheek rash.

The RegiSCAR score indicates no DRESS syndrome for this subject. SKLSI believes causality may be related to YKP3089 or phenobarbital.

9. Subject Number 4-19 in Phenobarbital Interaction Study

The subject is a 34 year old healthy Hispanic female volunteer who screened for the study YKP3089C022 "An Open-Label, Multiple-Dose, Pharmacokinetic Drug Interaction Study to Assess the Effects of YKP3089 on Phenobarbital in Healthy Adult Subjects" on 13 May 2015.

The subject had no significant medical history. All laboratory results obtained at screening were WNL, including LFTs negative virology for HIV, and negative for Hepatitis B and C. The subject had no prior history of similar rash and no other history of reactions to medications.

On 28 May 2015, the subject began dosing per protocol titration scheme, with at home dosing beginning Day 3 as follows:

- Day 1-7 Phenobarbital 30 mg tid day
- Day 8-22 Phenobarbital 90 mg q day
- Day 23-28 Phenobarbital 90 mg q day, YKP3089 50 mg q day
- Day 29-35 Phenobarbital 90 mg q day, YKP3089 100 mg q day
- Day 36-42 Phenobarbital 90 mg q day, YKP3089 150 mg q day
- Day 42-56 Phenobarbital 90 mg q day, YKP3089 200 mg q day
- Day 57-59 Phenobarbital 60 mg q day
- Day 60-62 Phenobarbital 30 mg q day

On Day 57, 23 July 2015, the subject stopped YKP3089 per the protocol titration scheme above and continued with phenobarbital. On Day 59, 25 July 2015, the subject reported development of a moderate generalized rash on the neck, chest, abdomen, arms, and back during the phenobarbital taper. 25 July 2015 is 37 days after starting YKP3089, 3 days after stopping YKP3089 per protocol and 59 days after starting phenobarbital. There was no swelling of the lips, tongue, or throat. Phenobarbital was discontinued on 28 July 2015. At this time, the rash was noted to be more maculopapular in appearance than from onset. The subject was advised to take 25-50 mg Diphenhydramine every 4-6 hours as needed, and was released from the clinic later that afternoon. All lab results obtained on Day 62, 28 July 2015, were WNL including eosinophils 400/uL, percent eosinophils 8.3%, and hepatic enzymes.

The subject called the nursing staff on the evening of 29 Jul 2015 to report that the rash appeared worse and had spread to her legs. The subject complained of numbness to her tongue, discomfort and pain with swallowing, and itchiness. The subject was evaluated on the following day, on 30 July 2015. She was afebrile with non-tender, mildly enlarged bilateral anterior cervical lymph nodes. An erythematous maculopapular rash to the trunk, neck, upper bilateral arms, and upper bilateral legs was noted during the skin exam. Prednisone was initiated at 60mg on 30 July 2015; reduced to 50mg on 1 Aug 2015, reduced to 40mg on 3 Aug 2015 and reduced to 20mg 4Aug 2015. The subject was reexamined on 4 August 2015 and the rash was found to extended to below the knees and distal to the elbows bilaterally. The rash it also appeared more erythematous and confluent to the chest/trunk. Prednisone was increased to 60mg on 4 Aug 2015 and tapered off until 19 Aug 2015 when it was stopped. The subject reported a new rash with pruritus to the bilateral feet on 14 September 2015. Coincidentally, the subject had an allergist appointment at the Mayo Clinic on 15 September 2015. During this

appointment, reportedly, she was diagnosed with “type 4 hypersensitivity reaction and type 4 skin reaction”. The subject was prescribed fluocinamide topical ointment and advised to use ibuprofen for the sore throat. All AEs resolved.

The RegiSCAR score indicates no DRESS syndrome for this subject. SKLSI believes the causality may possibly be related to YKP3089 or phenobarbital.